

WEB TECHNOLOGY

1. Concept of Web Technology

The word web describes how information, people, and devices are connected online. Web technology refers to the tools, software, protocols, and programming languages used to create, develop, publish, and manage applications and content on the Internet.

Examples of web technologies: HTML, CSS, JavaScript, PHP, Python, MySQL, and WordPress.

2. Web Development Life Cycle (WDLC)

The Web Development Life Cycle (WDLC) is a step-by-step process used to plan, design, develop, test, launch, and maintain a website or web application. It provides a roadmap for developers, designers, clients, and other stakeholders to collaborate and deliver a successful online solution.

Stage	Main activity
1. Gathering Information	Identify purpose, target users, requirements, goals, and available resources.
2. Planning	Prepare scope, site map, technologies, schedule, and project plan.
3. Designing and Layout	Design page structure, navigation, visual appearance, wireframes, and user interface.
4. Content Creation	Prepare text, images, graphics, videos, and other required content.
5. Development	Write and integrate HTML, CSS, JavaScript, server-side code, databases, and other technologies.
6. Testing	Check functionality, links, forms, compatibility, responsiveness, performance, and usability.
7. Maintenance and Update	Fix errors, improve security, update content, and add new features as required.

3. Important Areas and Trends in Web Development

Area	Short explanation
AI and Machine Learning Integration	Using intelligent systems to personalize services, automate tasks, analyze data, and improve web applications.
Progressive Web Apps (PWA)	Web applications designed to provide an app-like experience on supported devices.
Augmented Reality (AR) and Virtual Reality (VR)	Using immersive or interactive digital environments through the web.
Responsiveness	Designing websites so that they work properly on phones, tablets, laptops, and desktops.
Edge Computing	Processing some data closer to users or devices to reduce delay and improve performance.
Web Security	Protecting websites, applications, accounts, and data from unauthorized access and attacks.

4. Domain Name System (DNS)

DNS: Domain Name System.

DNS is a naming system that translates human-readable domain names, such as example.com, into IP addresses that computers use to locate servers on a network. It makes websites easier to access because users can remember names instead of numerical IP addresses.

4.1 Common Types of Domain Names

Domain	Common use
.com	Commercial websites and businesses
.net	Network-related organizations and services
.np	Nepal country-code domain
.edu	Educational institutions
.gov	Government organizations
.org	Organizations, often non-profit or community based

4.2 DNS / Domain Registration Process

1. Choose a suitable domain name.
2. Check whether the domain name is available.
3. Choose a reliable domain-name registrar.
4. Register or purchase the domain name through the registrar.
5. Keep the registration account and domain-management details safe; do not lose control of the domain name.

5. Concept of UI/UX

Front end: The part of a website or application that users see and interact with directly, such as pages, buttons, menus, forms, and visual elements.

Back end: The part of a web application that works behind the scenes, such as servers, databases, business logic, authentication, and APIs.

UI (User Interface): The visual and interactive part of a product through which users interact with the system.

UX (User Experience): The overall experience a user has while using a website, application, or digital product, including ease, usefulness, efficiency, and satisfaction.

5.1 Advantages of Good UI/UX Design

- Increases user engagement through attractive and clear design.
- Improves the overall user experience.
- Helps create a unique visual identity for branding.
- Makes websites and applications easier to use and navigate.
- Can reduce development time and cost by identifying problems early.

5.2 Software Used for UI/UX Design

Commonly used tools include Figma and Adobe XD. Other tools may also be used depending on the course or workflow.

Tool	Typical use
Figma	Interface design, wireframing, prototyping, and collaborative design.
Adobe XD	UI design and interactive prototyping.

Plasma	Listed in the source material as a UI/UX tool.
Mix	Listed in the source material as a UI/UX tool.

6. Wireframe

Wireframe: A simple visual plan or skeletal layout of a webpage or application that shows the structure, placement of content, navigation, and key interface elements before detailed design and coding.

6.1 Key Aspects of Wireframe Design

Aspect	Meaning
Structure / Framework	Shows the basic arrangement of page elements.
User Flow and Navigation	Shows how users move from one page, screen, or section to another.
Quick Interaction	Allows early checking of how users may interact with the layout.
Prototyping	Helps create an early model of the interface before final implementation.
Cost Efficiency	Helps identify layout problems early, reducing later redesign work.

6.2 Benefits of Wireframe

- Easy to draw
- Easy to understand
- Easy to modify
- No coding required

7. Introduction to HTML

HTML: HyperText Markup Language. It is the standard markup language used to structure content on web pages.

HTML uses tags and elements to tell a web browser how different pieces of content are organized, such as headings, paragraphs, links, images, lists, tables, and forms.

7.1 Short History of HTML

HTML was introduced in the early 1990s as a way to structure and connect documents on the World Wide Web. It has developed through several versions, with modern HTML5 providing many features for current web pages and applications.

7.2 Text Editors for HTML

- Sublime Text
- Visual Studio Code
- Notepad++

8. HTML Tags

HTML tags are keywords enclosed in angle brackets (< >) that define or mark up elements in an HTML document.

Type	Definition	Example
Container / Paired Tag	Has both an opening tag and a	Welcome

	closing tag, with content between them.	
Empty / Unpaired Tag	Does not require a separate closing tag.	

9. HTML Attributes

An HTML attribute provides additional information about an HTML element. It is normally written inside the opening tag as name="value".

```
<a href="https://example.com" target="_blank">Visit Website</a>
```

In the example, href and target are attributes of the anchor tag.

10. Basic Structure of an HTML Document

```
<!DOCTYPE html>
<html>
<head>
  <title>My Web Page</title>
</head>
<body>
  <h1>Welcome</h1>
  <p>This is my first webpage.</p>
</body>
</html>
```

Part	Purpose
DOCTYPE	Declares the document type.
<html>	Root element of the webpage.
<head>	Contains information about the document, such as title and metadata.
<title>	Sets the title shown in the browser tab.
<body>	Contains the visible webpage content.

11. Organizing Text in HTML

11.1 Comment Tag

HTML comments are notes written for developers. They are not displayed as normal page content in the browser.

```
<!-- This is a comment -->
```

11.2 Line Break Tag (
)

The
 tag inserts a line break without starting a new paragraph.

```
Hello<br>World
```

11.3 Paragraph Tag (<p>)

The <p> tag defines a paragraph of text.

```
<p>This is a paragraph.</p>
```

11.4 Aligning Text

In older HTML examples, the align attribute may be used with values such as left, center, or right. In modern HTML, CSS text-align is preferred.

Value	Meaning
left	Aligns content to the left.
center	Centers content.
right	Aligns content to the right.

12. Headings in HTML

HTML provides six heading levels from <h1> to <h6>. <h1> is the highest-level heading and <h6> is the lowest-level heading.

<h1>Main Heading</h1>

<h2>Subheading</h2>

<h3>Section Heading</h3>

13. Horizontal Rule (<hr>)

The <hr> element creates a thematic horizontal line or divider.

Attribute / setting	Purpose
align	Specifies alignment in older HTML usage.
size	Specifies line thickness in older HTML usage.
width	Specifies the width of the line.
color	Specifies the color in older HTML usage.

<hr align="center" size="3" width="60%" color="blue">

Note: Modern HTML/CSS normally uses CSS properties such as width, height, border, background, and margin instead of deprecated presentation attributes.

14. Text Styling in HTML

Tag / attribute	Purpose	Example
	Bold text	Bold
<i>	Italic text	<i>Italic</i>
<u>	Underlined text	<u>Underline</u>
<sup>	Superscript	x²
<sub>	Subscript	H₂O

Older HTML lessons may also demonstrate face, size, and color attributes with the tag. These are obsolete in modern HTML and are generally replaced with CSS.

Sample Text

14.1 Common HTML Color Names

Common named colors include:

black, red, blue, silver, gray, purple, white, green, yellow, lime, maroon, and aqua.

15. Marquee Tag

The <marquee> tag was historically used to create scrolling text or images. It is obsolete in modern HTML, but it is often included in introductory HTML textbooks.

```
<marquee>Welcome to my website</marquee>
```

16. Anchor Tag (<a>)

Anchor tag: The <a> tag is used to create a hyperlink to another webpage, document, location, email address, or other resource.

```
<a href="https://www.example.com">Visit Example</a>
```

16.1 Main Anchor Attributes

Attribute	Purpose
href	Specifies the destination of the link.
target	Specifies where the linked document should open.

16.2 Values of the target Attribute

Value	Meaning
_blank	Opens the link in a new browsing context, commonly a new tab.
_parent	Opens the link in the parent browsing context.
_self	Opens the link in the current browsing context.
_top	Opens the link in the top-level browsing context.

17. Hypertext

Hypertext is text containing links that allow users to move to another document, webpage, section, or resource.

Type	Meaning	Example
External Link	Links to a different website or external document.	External
Internal Link	Links to another location within the same webpage or website.	About

17.1 External Document Reference

```
<a href="notes.pdf">Open Notes PDF</a>
```

17.2 Intra-page Link / Bookmark

An intra-page link moves the user to a named location within the same page. Modern HTML commonly uses the id attribute as the target.

```
<a href="#contact">Go to Contact</a>
```

```
<h2 id="contact">Contact</h2>
```

Older HTML materials may refer to this as a name anchor.

```
<a name="contact"></a>
```

17.3 Email Link

`<a href="mailto:student@example.com">Email Me</a>`

18. HTML Lists

18.1 Unordered List

An unordered list displays items using bullets.

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

18.2 Ordered List

An ordered list displays items in a sequence, commonly using numbers.

```
<ol>
  <li>Plan</li>
  <li>Design</li>
  <li>Develop</li>
</ol>
```

19. Adding Images to HTML

The `<img>` tag is used to display an image. It is an empty element and normally does not need a closing tag.

Attribute	Purpose
src	Specifies the image file path or URL.
align	Specifies alignment in older HTML usage.
border	Specifies an image border in older HTML usage.
width	Sets image width.
height	Sets image height.
hspace	Specifies horizontal spacing in older HTML usage.
vspace	Specifies vertical spacing in older HTML usage.
alt	Provides alternative text when the image cannot be displayed and improves accessibility.

```

```

20. HTML Tables

An HTML table arranges information in rows and columns.

```

<table border="1">
  <tr>
    <th>Name</th>
    <th>Class</th>
  </tr>
  <tr>
    <td>Arom</td>
    <td>9</td>
  </tr>
</table>

```

20.1 Common Table Attributes / Concepts

Attribute / concept	Purpose
border	Specifies table border in traditional HTML usage.
bbgcolor	Sets the background color in older HTML usage; CSS is preferred today.
height	Sets table height.
width	Sets table width.
align	Aligns the table in older HTML usage.
cellpadding	Space between cell content and its border.
cellspacing	Space between table cells in older HTML usage.
colspan	Makes a cell span multiple columns.
rowspan	Makes a cell span multiple rows.
background	Specifies a background image in older HTML usage; CSS is preferred.

20.2 Caption Tag

The <caption> tag gives a title or caption to a table.

```

<table border="1">
  <caption>Student Details</caption>
  <tr><th>Name</th><th>Marks</th></tr>
  <tr><td>Rita</td><td>85</td></tr>
</table>

```

20.3 valign and align

Attribute	Value	Meaning
valign	top	Aligns cell content toward the top.
valign	middle	Aligns cell content vertically in the middle.
valign	bottom	Aligns cell content toward the bottom.
align	left	Aligns content to the left.
align	center	Centers content.
align	right	Aligns content to the right.

```

<td valign="middle" align="center">80</td>

```

These presentation attributes are common in older HTML lessons. Modern web development normally uses CSS for alignment.

21. HTML Form

The <form> element is used to collect information entered by users and submit it for processing.

21.1 Common Form Elements

Element	Purpose / example
Text field	Used for short text input. Example: <input type="text" name="username" size="30">
Password field	Used to enter passwords while hiding the typed characters. Example: <input type="password" name="pwd">
Textarea	Used for multi-line text. Example: <textarea name="message" rows="4" cols="30"></textarea>
Radio button	Allows the user to select one option from a group with the same name.
Checkbox	Allows the user to select zero, one, or multiple options.
Select box	Provides a drop-down list of options using <select> and <option>.
Submit button	Sends the form data for processing.
Reset button	Restores form fields to their initial values.

21.2 Text Field

The text field is a single-line input control. The type, name, and size attributes are commonly used in basic HTML examples.

```
<input type="text" name="fullname" size="30">
```

21.3 Textarea

A textarea provides a multi-line input area. Common attributes include name, rows, and cols.

```
<textarea name="message" rows="5" cols="40"></textarea>
```

21.4 Radio Buttons

```
<input type="radio" name="gender" value="male"> Male
```

```
<input type="radio" name="gender" value="female"> Female
```

21.5 Checkboxes

```
<input type="checkbox" name="skill" value="html"> HTML
```

```
<input type="checkbox" name="skill" value="css"> CSS
```

21.6 Select Box

```
<select name="class">
```

```
<option>Grade 8</option>
```

```
<option>Grade 9</option>
```

```
<option>Grade 10</option>
```

```
</select>
```

21.7 Submit and Reset Buttons

```
<input type="submit" value="Submit">
```

```
<input type="reset" value="Reset">
```

22. HTML <div> Tag

Division tag: The <div> element is a block-level container used to group related content and create sections of a webpage.

22.1 Advantages / Uses of the <div> Tag

- Grouping content into meaningful sections.
- Helping create structured and responsive page layouts.
- Connecting grouped content with JavaScript.
- Organizing page layout and structure.
- Applying common CSS styling to a whole section.

```
<div class="box">
  <h2>About School</h2>
  <p>Our school provides quality education.</p>
</div>
```

23. Dynamic HyperText Markup Language (DHTML)

DHTML: Dynamic HyperText Markup Language. It is a combination of HTML, CSS, JavaScript, and the Document Object Model (DOM) used to create webpages whose content or appearance can change dynamically in response to user actions.

24. Cascading Style Sheets (CSS)

CSS: Cascading Style Sheets. CSS is used to control the appearance, layout, spacing, colors, fonts, and responsive presentation of HTML documents.

24.1 How CSS Works

A CSS rule selects one or more HTML elements and applies properties with specified values to those elements.

```
p {
  color: blue;
  font-size: 18px;
}
```

In this example, p is the selector, color and font-size are properties, and blue and 18px are values.

25. CSS Properties and Values

```
selector { property: value; }
```

Property: The style characteristic being changed, such as color, font-size, or margin.

Value: The setting assigned to a property, such as blue, 16px, or 10px.

26. CSS Selectors

Selector	Meaning	Example
Element selector	Selects all elements of a particular HTML tag.	p { color: blue; }
ID selector	Selects the element with a specific id.	#header { color: red; }
Class selector	Selects elements that have a specified class.	.box { padding: 10px; }
Universal selector	Selects all elements.	* { margin: 0; }

27. Types of CSS

27.1 Inline CSS

Inline CSS is written directly inside the opening HTML tag using the style attribute.

```
<p style="color: red;">Red text</p>
```

27.2 Internal CSS

Internal CSS is written inside a <style> element in the <head> section of an HTML document.

```
<style>
p { color: blue; }
</style>
```

27.3 External CSS

External CSS is stored in a separate .css file and linked to the HTML document.

```
<link rel="stylesheet" href="style.css">
```

28. Common CSS Properties

Property	Typical value / example	Use
color	red / #333	Sets text color.
text-align	left / center / right	Aligns text.
text-decoration	underline / none	Adds or removes text decoration.
text-transform	uppercase / lowercase / capitalize	Changes letter case.
line-height	1.5 / 24px	Controls vertical space between lines.
font-family	Arial, sans-serif	Sets the typeface.
font-size	16px	Sets text size.
font-style	normal / italic	Sets the font style.
font-weight	normal / bold / 700	Controls text thickness.
background-color	lightblue	Sets background color.
background-image	url("bg.jpg")	Sets a background image.
background-repeat	repeat / no-repeat	Controls background-image repetition.
background-position	center / top right	Controls the position of a background image.
border-style	solid / dashed / dotted	Sets border line style.
border-width	1px / 2px	Sets border thickness.
border-color	black	Sets border color.

29. CSS Box Model

CSS box model: A model that treats each HTML element as a rectangular box made up of content, padding, border, and margin. It is important for understanding spacing and layout.

29.1 Main Parts of the Box Model

Part	Meaning
Content	The actual text, image, or other element content inside the box.
Padding	Space between the content and the border.
Border	The visible line surrounding the padding and content.
Margin	Space outside the border that separates the element from other elements.

```
div {
  width: 300px;    /* content width */
  padding: 20px;   /* inner space */
  border: 2px solid black;
  margin: 10px;    /* outer space */
}
```

A simple way to remember the order is: Content → Padding → Border → Margin.

30. CSS Measurement Units

Unit	Meaning	Example
em	Relative unit based on the font size of the element or its relevant parent context.	font-size: 2em;
px	Pixel; a commonly used screen-oriented unit for fixed-size values.	width: 300px;
mm	Millimetre; an absolute physical unit, mainly useful for print-oriented measurements.	width: 100mm;